Vijay Mohanram Iyer

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SUMMARY

Graduate engineer specializing in computer vision, machine learning, and embedded systems for intelligent safety and health applications. Experienced in developing and fine-tuning deep learning architectures, transformer models, and end-to-end pipelines. Skilled in optimization of dynamic systems, signal processing, and deploying efficient models on embedded platforms for real-time, user-centered solutions.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, JavaScript, SQL, Node.js, HTML, CSS
- **Libraries/Frameworks:** OpenCV, Qt, TensorFlow, PyTorch, pandas, NumPy, Matplotlib, scikit-learn, React.js
- **ADAS:** Sensor fusion, camera and LiDAR calibration, Point cloud, object detection & tracking
- **Embedded Systems:** ESP32, STM32, Arduino, Raspberry Pi, RTOS, Embedded C/C++
- **Tools & Cloud Platforms:** Git, Docker, MATLAB, Linux, Carla, conda, LaTeX, Jupyter Notebook, Azure, AWS

CORE COMPETENCIES

- **Technical Leadership:** Project Development, Cross-functional Collaboration, Innovative mindset.
- **Problem Solving:** Critical Thinking, Performance Optimization, Debugging.
- **Communication:** Active Listening, Documenting, Knowledge Transfer.
- **Spoken Languages:** English C2, German A2 (Learning)

PROFESSIONAL EXPERIENCE

1. Forschungszentrum Informatik — *September 2025 – Present*
   Research Assistant
   - Evaluate and optmize detection models through transfer learning and ensemble methods for hyperparameter fine-tuning and multimodal data analysis.
   - To utilize 3D geometric facemesh for reconstruction of desired regions.
   - Exploring GAN based approaches and diffusion techniques and their architectures for video rendering .
   - Conduct literature review and work towards research publication.
   - **Technology:** *Python, PyTorch, Gen AI, OpenCV*
2. TecoLab — *March 2023 – Present*
   Working Student
   - Sensor optimization for open-earables with TinyML and quantized neural networks for on-device inference.
   - ML4Print: Designed a CNN-based pipeline to classify printing techniques and paper substrates, using saliency maps for explainability and reducing manual forensic effort by 80%.
   - Applied regression models and data integration to simulate liquid thermal behavior in industrial valves, improving predictive maintenance and system reliability.
   - **Technology:** *Python, SciPy, pandas, NumPy, Arduino, Edge ML*
3. Forschungszentrum Informatik — *February 2025 – August 2025*
   Master Thesis (Grade: 1,0)
   - Developed an end-to-end deep learning pipeline using rPPG signals for artifact evaluation in vital signals and benchmarked under real-world conditions.
   - Designed a Fusion Model combining Vision Transformer and CNN features via a custom Fusion-Head for DeepFake classification.
   - **Technology:** *Python, PyTorch, Butterworth Filters, pandas, NumPy, Docker, Linux, conda, LaTeX*

4. Access@KIT *March 2023 – May 2023*
Working Student

- Implemented a diffusion-based text-to-speech framework leveraging phoneme alignment and acoustic features to generate natural, high-quality speech.
- Refined the speech synthesis pipeline via data preprocessing and feature extraction, delivering clearer and more accessible speech for end users.

– **Technology:** *Python, PyTorch, Diffusion Models, TTS, Audio Processing, React.js, MUI*

5. Accenture India Pvt. Ltd. *February 2022 – April 2022*
Associate Software Engineer

- Maintained IBM Mainframe servers with batch job scheduling and performance monitoring.

– **Technology:** *COBOL, Batch Processing, System Monitoring*

6. Accur Digitus *June 2021 – January 2022*
Web Developer Intern

- Developed responsive web applications using React.js and Tailwind CSS, integrated scalable RESTful APIs, and enhanced user experience, resulting in over 30% increase in engagement.

– **Technology:** *React.js, Tailwind CSS, REST APIs, JavaScript, Node.js*

EDUCATION

M.Sc. Electrical Engineering and Information Technology *May 2022 – September 2025*
Karlsruhe Institute of Technology
GPA: 2.3

Bachelor of Engineering in Electronics Engineering *August 2017 – May 2021*
University of Mumbai, India
GPA: 2.8

PROJECTS

1. **Non-Contact based Vital Extraction for Health Monitoring with Attention Mechanisms (FZI)**
   Utilized Plane-Orthogonal-to-skin and Wavelet Transform to extract pulsatile components from facial videos, improving robustness for non-contact vital sign monitoring in telemedicine.
   Technology: *Python, PyTorch, OpenCV, NumPy, Matplotlib, ViT, GenAI*

2. **CamCussion (Zeiss Innovation Hub)**
   Utilized pupil segmentation, gaze tracking, and saccadic velocity estimation to analyze pupil dilation and eye movements in real time, contributing to accurate concussion diagnosis.
   Technology: *Python, OpenCV, Dlib, PyQt, Image Processing, PyTorch, NumPy, ML*

3. **Real-Time 3D Obstacle Detection Using Raw LiDAR Point Cloud Data**
   Developed real-time 3D obstacle detection by voxelizing LiDAR point clouds into ViT tokens and applying multi-head attention for improved accuracy.
   Technology: *Python, PyTorch, Open3D, NumPy, ViT, LiDAR Processing*

4. **Real-Time Car Accident Alert System**
   Built ESP32-based embedded LSTM system with TensorFlow Lite for real-time crash detection and automated SMS/email alerts with GPS location.
   Technology: *Python, ESP32, Deep Learning, Embedded C*